

Goutham Raju Kosuru Srinivasa

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EDUCATION

Arizona State University, Tempe, AZ

Jan. 2024 – Dec. 2025

Master of Science in Information Technology — With Distinction, GPA 4.0/4.0

TECHNICAL SKILLS

Languages: C#, Java, Python, TypeScript, JavaScript, HTML, CSS, SQL

Technologies and Frameworks: .NET Core, ASP.NET Core Web API, FastAPI, NestJS, Angular, React.js, Next.js

Developer Tools: VS Code, Git, GitHub Actions, Postman, Docker

Cloud and DevOps: AWS (Lambda, API Gateway, DynamoDB, S3, SQS, Bedrock, SAM, CloudFormation, EC2), Azure (App Services, DevOps, Application Insights)

Database and Storage: PostgreSQL, SQL Server, DynamoDB, MongoDB, Query Optimization, Schema Design

AI / LLM Tools and Agents: LangChain, AWS Bedrock, Claude Code, GitHub Copilot, Cursor

EXPERIENCE

Community Dreams Foundation – Software Engineer (Volunteer)

Feb. 2026 – Present

Tech: Python, FastAPI, PostgreSQL, C#, React, Docker, GitHub Actions

Remote

- Rebuilt a compliance-critical platform from scratch after finding the existing codebase was entirely hardcoded UI with no functional backend; designed a PostgreSQL schema with enforced state transitions and row-level security.
- Implemented RBAC with AES-256 encrypted storage and immutable audit trails, enforcing access boundaries at the database level.
- Established a 26-test automated suite validating data integrity and concurrent state transitions, integrated into a CI/CD pipeline via Docker and GitHub Actions, all passing before feature work began.
- Built structured logging and observability dashboards before the first user onboarded, enabling proactive detection of data integrity issues via state machine trace analysis; onboarded **20+ beta users** across three distinct roles.

Neudesic, an IBM Company — Software Development Engineer

Feb. 2022 – Jan. 2024

Tech: C#, .NET Core, Angular, TypeScript, SQL Server, Azure Application Insights, Azure DevOps

Bangalore, India

- Delivered three connected web and mobile applications for SLB (Schlumberger), a global oilfield services company, replacing manual phone-based booking with a customer portal, employee app, and admin dashboard synced in real time, serving **2,000+ daily users**.
- Diagnosed a high-impact SQL performance defect using SQL Server Profiler, identifying N+1 query patterns firing sequential calls per request; rewrote the data access layer with batched JOINS and composite indexes, cutting API latency by **32%**.
- Led a TypeScript migration across all three codebases, eliminating a recurring class of null-reference errors caught only in production; reduced runtime errors by **40%**.
- Established production monitoring using Azure Application Insights, diagnosing and resolving incidents end-to-end; improved CI/CD pipeline reliability by **25%** through root cause analysis of recurring build failures, maintaining **99.5% uptime**.

The Sparks Foundation — Software Engineer Intern

Aug. 2021 – Oct. 2021

Tech: Python, AWS Lambda, API Gateway, DynamoDB, SQS, SAM, CloudFormation

Remote

- Built a serverless, event-driven backend on AWS Lambda and API Gateway, designing a DynamoDB schema with access-pattern-first modeling and GSIs to sustain **1,000+ daily transactions at sub-300ms p95 latency**.
- Automated infrastructure provisioning using AWS SAM and CloudFormation, reducing manual environment setup from 45+ minutes to under 10 minutes, cutting provisioning time by **80%**.

PROJECTS

Multi-Agent AI Document Processing Pipeline — Tech: AWS Bedrock, LangChain, FastAPI, Python, SQS, DynamoDB

- Architected a fault-tolerant, event-driven pipeline with three specialized agents (classify → extract → validate) orchestrated via SQS with dead letter queues and exponential backoff retry, ensuring failed messages never block downstream processing.
- Implemented idempotency checks so redelivered messages under SQS's at-least-once guarantee never process a document twice; used DynamoDB for agent state and audit persistence.

Secure Task Management System — Tech: NestJS, Angular, TypeScript, PostgreSQL, JWT, Docker, Nx

- Built a production-ready full-stack application in an Nx monorepo with JWT authentication, database-enforced three-role RBAC, and a shared TypeScript DTO library eliminating contract drift between frontend and backend.
- Designed a multi-tenant PostgreSQL schema with row-level data isolation and comprehensive automated test coverage, deployed via a Docker-based CI/CD pipeline.

PUBLICATION

Object Detection and Text Reading Device for the Visually Impaired

IRJMETS, 2022

Co-authored peer-reviewed research (Impact Factor: 6.752) on a real-time YOLOv3 and Tesseract OCR system deployed on Raspberry Pi 4, achieving 15+ FPS via TensorFlow Lite quantization.